

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

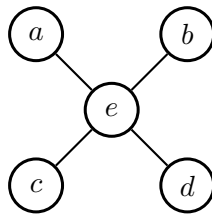
2023/24 Semester 2

MH1301 Discrete Mathematics

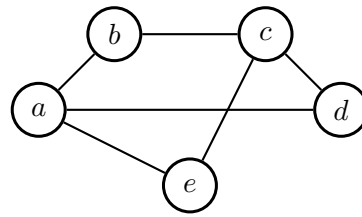
Tutorial 9

For the tutorial in Week 10 on 28 March, let us discuss the following questions.

Question 1. (Ex. 10.2.17, 18.) Determine whether the graph is bipartite.



Ex. 10.2.17

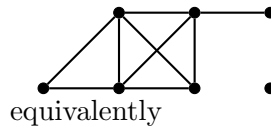


Ex. 10.2.18.

Question 2. (Ex. 10.2.22). For which values of n are these graphs bipartite?

- (a) K_n ,
- (b) C_n .

Question 3. (Ex. 10.2.27). The degree sequence of a graph is the sequence of the degrees of the vertices of the graph in nonincreasing order. For example, the degree sequence of the graph below is 4,4,4,3,2,1,0.



equivalently

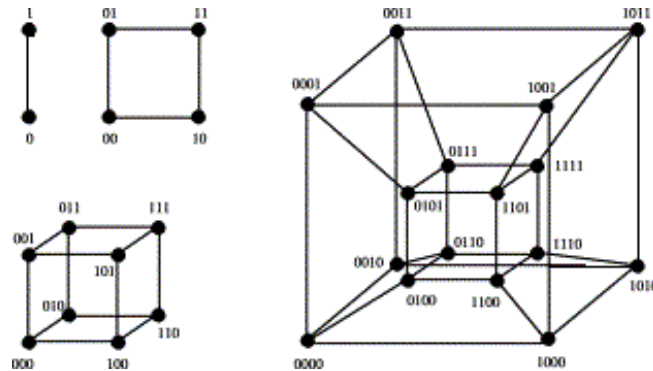
What is the degree sequence of the bipartite graph $K_{2,3}$? How about $K_{m,n}$?

Question 4. (Ex. 10.7.15). Prove that removing one edge from $K_{3,3}$ leads to a planar graph. How about K_5 ?

Question 5. The n -dimensional *hypercube* Q_n is a graph whose vertices are the 2^n strings of length n over the alphabet $\{0, 1\}$, and for which two vertices are connected if and only if the associated strings differ in exactly one position. Note that Q_n has 2^n vertices and $n/2 \cdot 2^n$ edges (the degree of all vertices is n).

The hypercubes Q_1, Q_2, Q_3, Q_4 ¹ look like this:

¹ Q_2 is a square, Q_3 a cube, and Q_4 is called a *tesseract*.



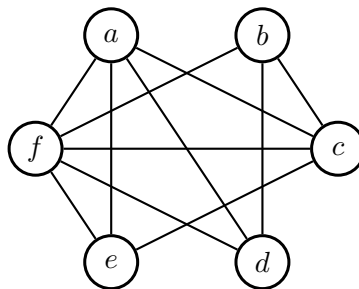
Determine the chromatic number of Q_n for all n .

Question 6. (Ex. 10.2.60. and Ex. 10.2.65).

- Find a planar graph G such that $\chi(G) = 4$.
- Prove that if, for a graph G , we remove one vertex and the edges incident to it, then the resulting graph G' has chromatic number either $\chi(G)$ or $\chi(G) - 1$.

Question 7. Ex. 10.7.1. Can five houses be connected to two utilities without connections crossing?

Question 8. Ex. 10.7.5. Determine whether the given graph is planar. If so, draw it so that no edge cross.



Question 9.

- Is there a planar simple connected graph with more than 3 vertices, such that removing any vertex and incident edges from it results in $\chi(G) - 1$? If so, find one.
- Is there a planar simple connected graph with 5 or less vertices, such that removing any vertex and incident edges from it results in $\chi(G)$? If so, find one.

Question 10.

- How many edges are on a Hamilton path?
- Is there a connected graph with an Euler path that has the degree sequence 4,4,2,1,1?

Answer.

- 1) Yes; Yes.
- 2) 1 and 2; all even n .
- 3) 2,2,2,3,3; $m, \dots, m, n, \dots, n$. (m many m 's followed by n many n 's.)
- 5) 2.
- 7) Yes.
- 8) Yes.
- 10) $n - 1$; No.